09. Business Interpretation, Results & Limitations

Turn cluster statistics into testable business hypotheses without overstating the evidence.

Project: SmartCart Customer Segmentation

Source checked: Sai-Likhith-2005/Customer-Segmentation | smartcart.ipynb | smartcart_customers.csv

Possible segment narratives

Cluster	Evidence-based descriptive label	Potential experiment
0	Lower-income, lower-spend, partner households; lower channel activity	Test low-cost relevant bundles or a carefully timed reactivation offer
1	Higher-income, high-spend, partner households; strong multi-channel purchasing	Test premium assortment, loyalty benefits or catalogue-led cross-sell
2	Lower-income, lowest-spend, mostly-alone households; high web visits relative to purchases	Test clearer product discovery, value messaging or friction reduction
3	Higher-income, high-spend, mostly-alone households; strongest observed campaign response	Test tailored premium campaigns, while monitoring contact fatigue

How to speak about the segments

These are analyst-created descriptions based on average profiles, not labels supplied by the business. Cluster 0 and 2 are both relatively low-spend; cluster 1 and 3 are both high-spend. The clearest differences between each pair involve household composition, channel behaviour and observed response rate.

Key limitations

- Response is included as a feature, so the high response rate in cluster 3 is not independent validation of the segment's future responsiveness.
- Age is calculated using the hard-coded year 2026 and should be tied to a fixed analysis date.
- All preprocessing and PCA are fit on the full dataset; there is no holdout validation.
- The 3D PCA space captures only 44.95% of variance.
- The notebook drops product-level spending columns after summing them, losing basket mix.
- Hard filters remove four records without documented domain justification.
- No explicit method is provided to assign future customers to Agglomerative clusters.
- The dataset is a snapshot and may not reflect current behaviour or market conditions.

Business validation plan

- Agree on an actionable decision for each segment before activation.
- Run randomized holdout tests within segments to estimate incremental lift, not just response rate.
- Compare conversion, revenue per contact, margin, unsubscribe/complaint rate and long-term retention.
- Monitor segment size and profile drift over time; refresh clustering only under a documented governance process.
- Check whether segment differences are robust across demographic and household subgroups.

Interview-ready conclusion

“The four-cluster solution suggests two broad spending tiers, with household and channel differences within each tier. Cluster 3 has the highest historical response rate, but Response was itself included in clustering, so I would not claim that the segment predicts future campaign success. I would exclude Response for prospective segmentation and validate campaign strategies with randomized experiments.”